

ECON 108

INTRODUCTION TO ECONOMICS

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Scarcity And the world of Trade-Offs

CHAPTER II

CHAPTER OUTLINE

Scarcity

Scarcity, Choice, and Opportunity Cost

The World of Trade-Offs

The Choices Society Faces

CHAPTER OUTLINE

Trade-off Between Present and Future Consumption

Comparative Advantage and Trade

SCARCITY

Occurs when the ingredients (resources) for producing things that people desire are insufficient to satisfy **all** wants.

SCARCITY

What scarcity is NOT:

- It is *not* a shortage
- It is *not* the same thing as poverty

SCARCITY

Production

- Any activity that results in the conversion of **resources** into products that can be used in consumption.

Resources or Factors of Production

- Inputs that are used to produce things that people want.

SCARCITY

Resources or Factors of Production

- Land
 - Natural resources or the “gifts of nature.”
- Labor
 - The human resource.

SCARCITY

Resources or Factors of Production

- **Physical Capital**
 - Factories and equipment used in production.
- **Human Capital**
 - Accumulated training and education of workers.

SCARCITY

Resources or Factors of Production

- Entrepreneurship
 - Person who organizes, manages, and assembles the other resources.
 - Risk taker.
 - Maker of basic business policy decisions.

SCARCITY

Needs (difficult to define)

- Life-saving necessity

Wants

- Desirable things that people wish to have
- People have unlimited wants

SCARCITY

In reality, every individual has competing wants but cannot satisfy all of them (Why?) Each person must therefore make choices.

Whenever a choice is made to produce or buy something, something else that is also desired is not produced or not purchased

Therefore some other wants will remain unsatisfied

Test yourself

Scarcity arises because of:

- a. Resources are finite and unable to meet all human wants and needs
- b. A natural disaster
- c. International companies are slow to explore for new resources
- d. All of the above

Test yourself

Which of the following is NOT a factor of production?

- a. Money
- b. Low-skilled labor
- c. Office telephones
- d. A registered nurse

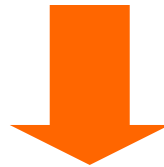
Test yourself

Rich people in Argentina can afford three hundred times more than poor people. Thus, the problem of scarcity is more severe among the poor.

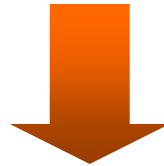
Is this statement true or false? Please justify your answer.

SCARCITY, CHOICE, AND OPPORTUNITY COST

Limited Resources & Unlimited Wants



Scarcity



Choices



Opportunity Cost

SCARCITY, CHOICE, AND OPPORTUNITY COST

Opportunity Cost

The highest-valued, next-best alternative that must be sacrificed to attain something or satisfy a want.

SCARCITY, CHOICE, AND OPPORTUNITY COST

In economics, cost is always a forgone opportunity.

SCARCITY, CHOICE, AND OPPORTUNITY COST

Some examples...

- What is the opportunity cost of attending this economics class?
- What is the opportunity cost of having a girlfriend/boyfriend?
- What is the opportunity cost of increasing research for an AIDS vaccine?

An example

Mike is going back to school full-time to finish his degree. He is currently employed and earns \$45,000 per year (after taxes). Tuition costs \$4,800 per year. Fees, supplies and other expenses amount to \$2,400 per year. The TOTAL cost of going back to school is:

- a. \$45,000 b. \$47,400 c. \$52,200 d. \$7,200

THE WORLD OF TRADE-OFFS

Whenever resources are used in some activity, the user is **trading off** the opportunity to use these resources for other things.

The value of this trade-off is represented by the **opportunity cost**.

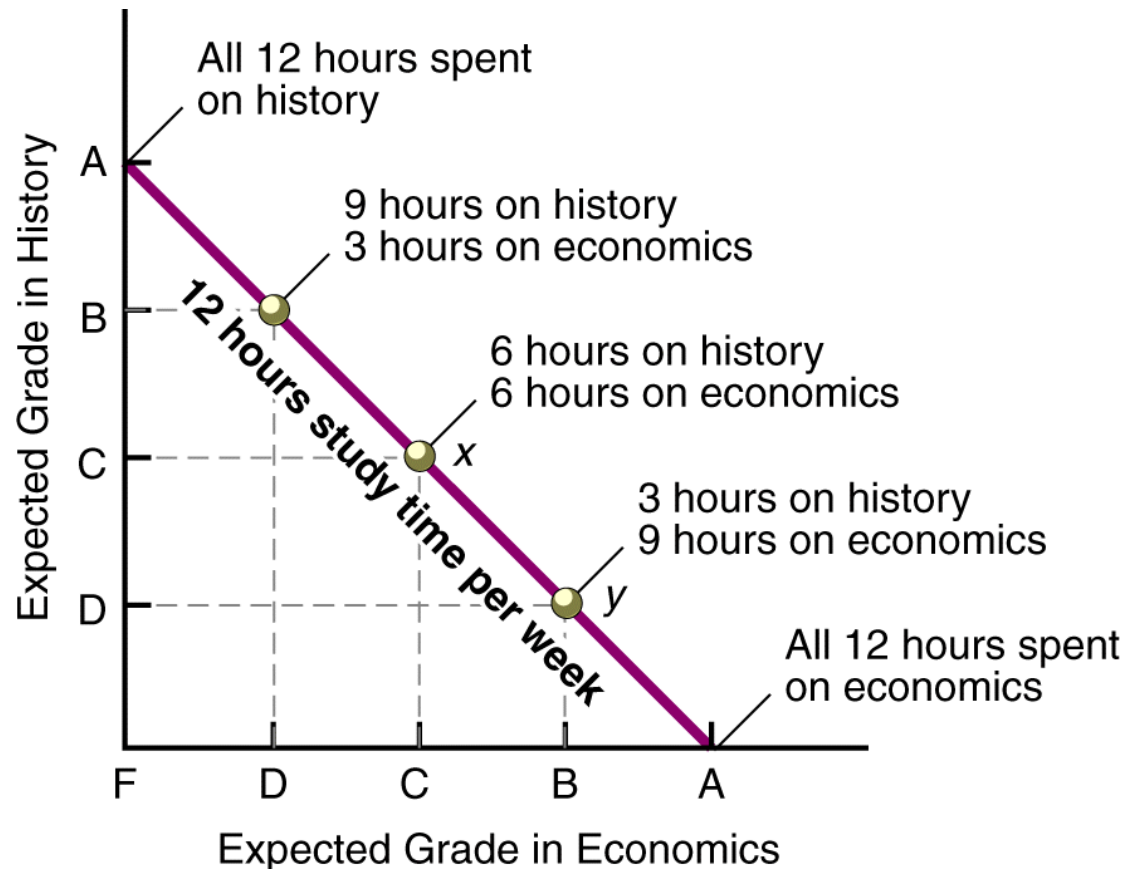
The opportunity cost of a choice is the highest-valued, next-best alternative that must be sacrificed.

THE WORLD OF TRADE-OFFS

The production possibilities curve (PPC)

- The PPC illustrates the combination of outputs that a society can produce if all of its resources are being used efficiently.

THE WORLD OF TRADE-OFFS



THE WORLD OF TRADE-OFFS

The production possibilities curve (PPC)

- Represents all possible combinations of outputs that can be produced assuming:
 1. a fixed amount of productive resources of a given quality,
 2. a given technology,
 3. and the efficient use of those resources.

THE WORLD OF TRADE-OFFS

In other words:

- The PPC is defined as the maximum quantity of one commodity that can be produced given that a specific quantity of another is produced, when all resources are efficiently employed.

THE WORLD OF TRADE-OFFS

Questions:

- What would happen to the production possibilities curve if you could spend more time studying?
- What would happen to the potential grades?
- Is the trade-off constant?

THE CHOICES SOCIETY FACES

Production Trade-Offs

- We are going to use the PPC to analyze the production trade-off facing the economy.
- To simplify the problem we assume the economy produces only two commodities.

THE CHOICES SOCIETY FACES

The production possibilities curve (PPC)

Assumptions:

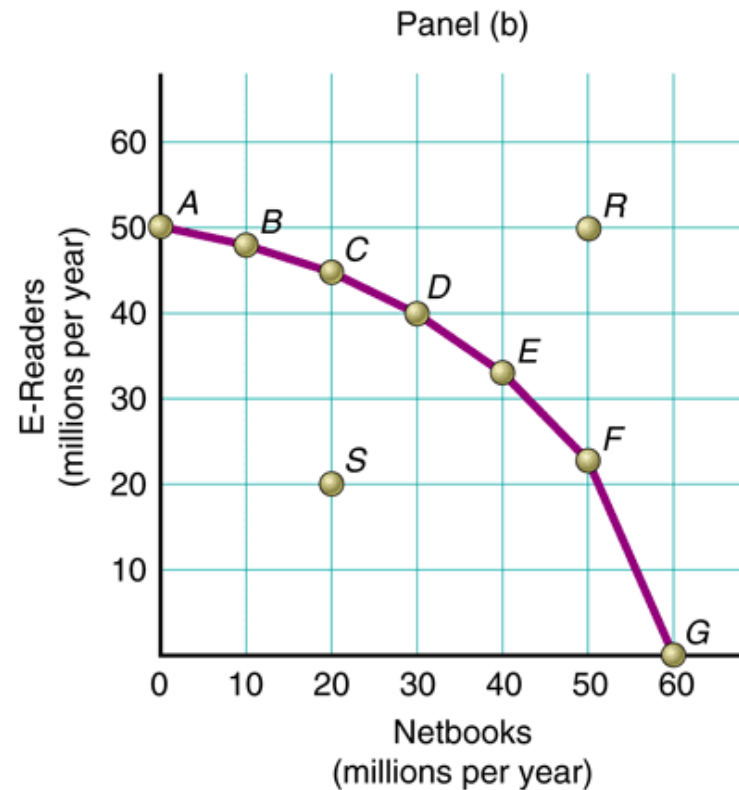
- Production is for a specific time period.
- Resources are fixed for the time period under consideration.
- Technology does not change over the time period.

THE CHOICES SOCIETY FACES

Trade-Off between E-readers and Netbooks

Panel (a)

Combination	E-Readers (millions per year)	Netbooks (millions per year)
A	50.0	0
B	48.0	10
C	45.0	20
D	40.0	30
E	33.0	40
F	22.5	50
G	0.0	60



THE CHOICES SOCIETY FACES

Two important relationships:

1. Relationship between opportunity cost and PPC.
2. Law of Increasing Additional Costs and PPC.

THE CHOICES SOCIETY FACES

Law of Increasing Additional Cost:

The opportunity cost of additional units of goods generally increases as society attempts to produce more of that good.

Why?

Because of specialized resources.

Test yourself

Define the following concepts:

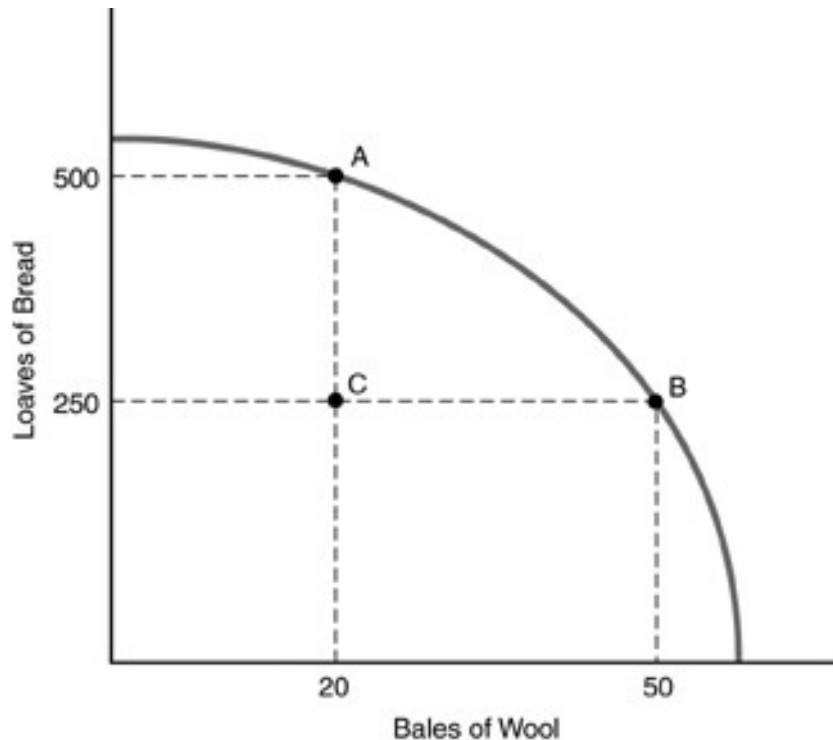
1. Opportunity Cost.
2. Production Possibilities Curve.
3. Law of Increasing Additional Costs.

Test yourself

Which of the following is held constant when constructing a production possibilities curve?

- A) the price level
- B) the income level
- C) combination of goods produced
- D) the amount of total resources used

Test yourself

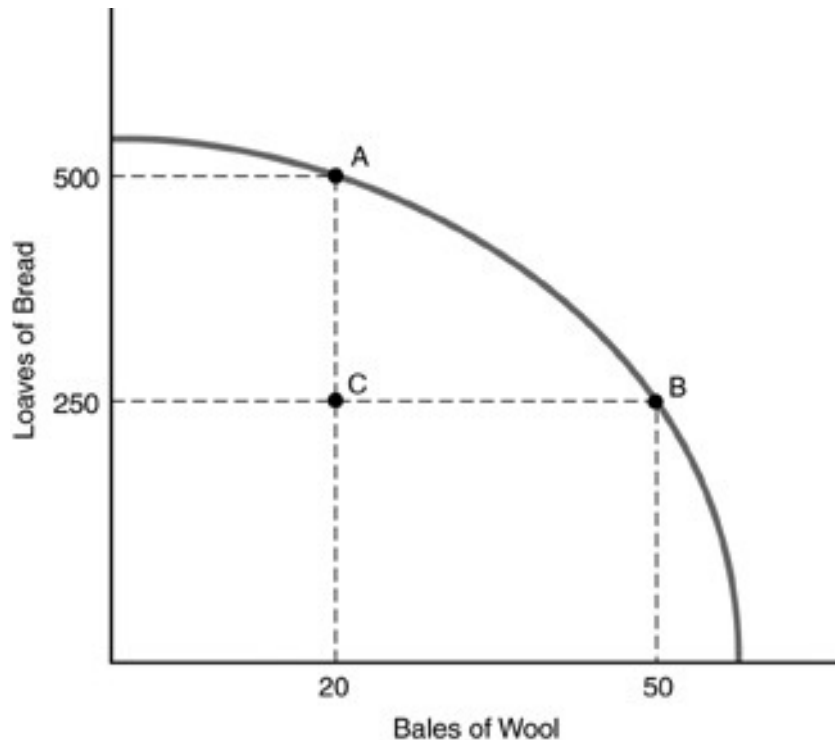


Production at Point C

- A) would indicate that this economy is producing beyond its capabilities.
- B) would indicate production at a level below that which is attainable.
- C) is not attainable with given resources and technology.
- D) would demonstrate the existence of scarcity.

Test yourself

Which one of the following statements is true:



A) Point C cannot be produced.

B) The best production point is 500 loaves of bread and 50 bales of wool.

C) The total amount of resources it takes to produce 20 bales of wool and 500 loaves of bread is more than the amount of resources needed to produce 50 bales of wool and 250 loaves of bread.

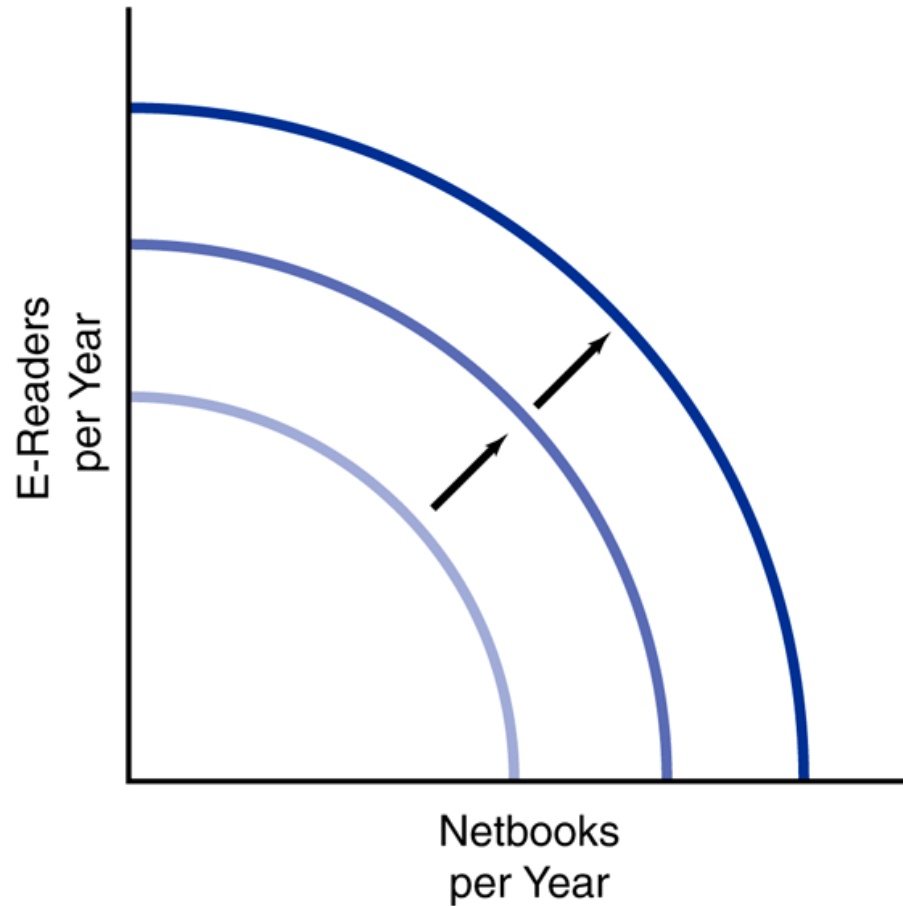
D) The total amount of resources it takes to produce 20 bales of wool and 500 loaves of bread is the same as the amount of resources needed to produce 50 bales of wool and 250 loaves of bread.

Economic Growth and the Production Possibilities Curve

Economic growth

- Increases the production possibilities of e-readers and netbooks
- *Over time*, it is possible to have more of everything
- Illustrated by an outward shift of the production possibilities curve

Economic Growth and the Production Possibilities Curve



The Trade-Off Between the Present and the Future

The PPC can be used to illustrate the trade-off between *present* and *future* consumption

Consumption

- The use of goods and services for personal satisfaction

Economic Growth and the Production Possibilities Curve



Consumer goods

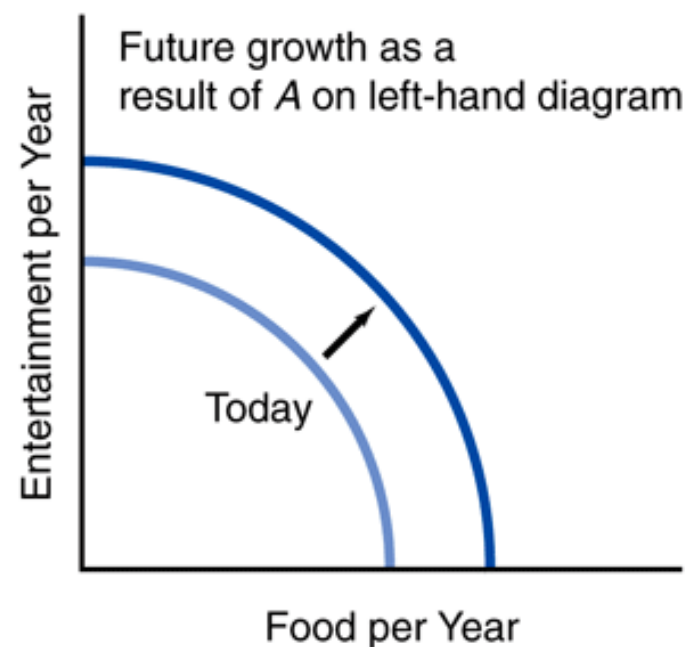
- Goods produced for personal satisfaction

Capital goods

- Goods used to produce other goods

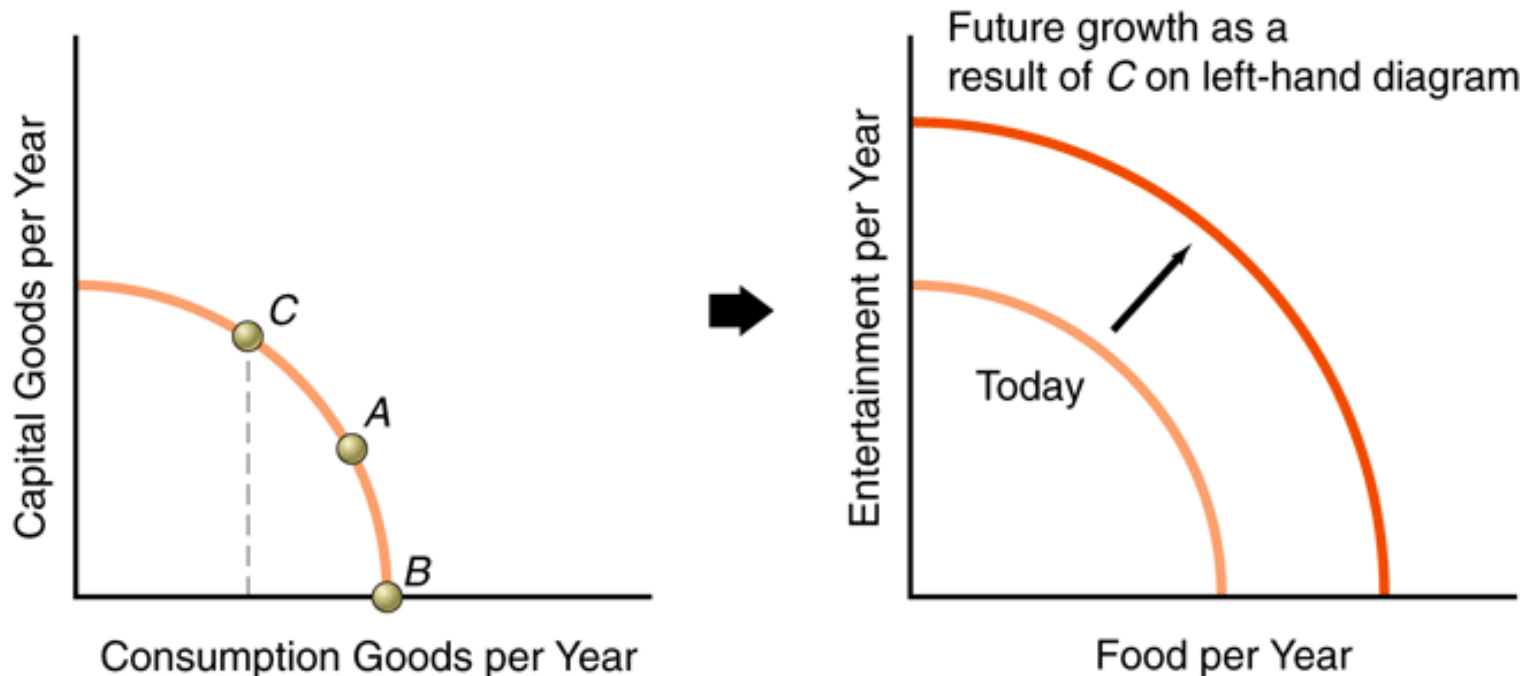
Economic Growth and the Production Possibilities Curve

Panel (a)



Economic Growth and the Production Possibilities Curve

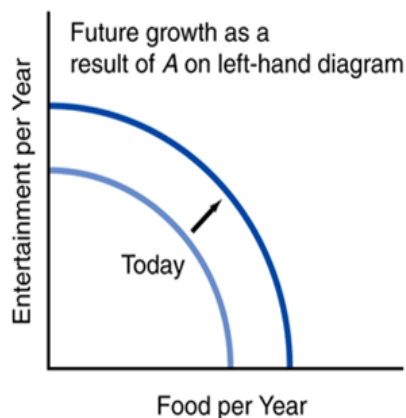
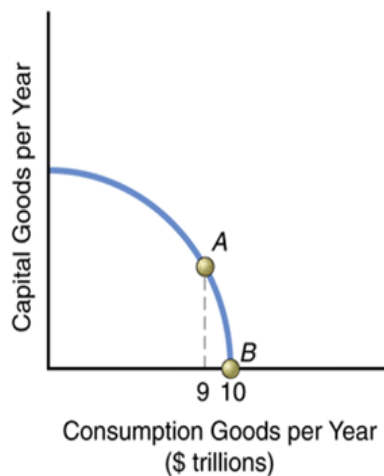
Panel (b)



Economic Growth and the Production Possibilities Curve

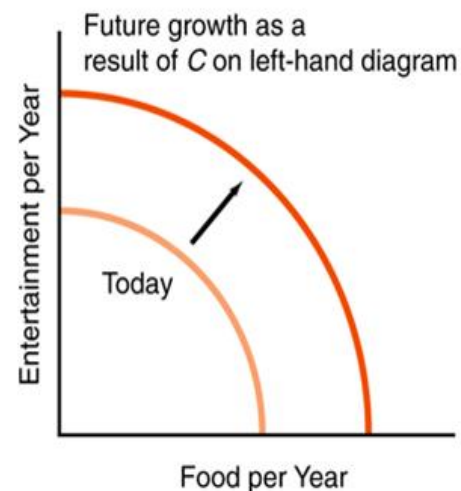
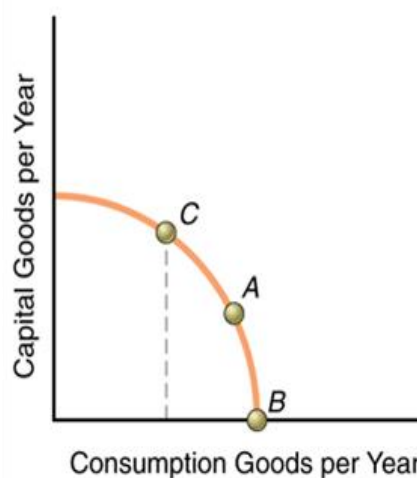
LOW INVESTMENT

Panel (a)



HIGH INVESTMENT

Panel (b)



Economic Growth and the Production Possibilities Curve

Capital Goods and Growth: Observations

- Forgo consumption goods to produce capital goods
- Increase in capital goods stimulates economic growth

The Trade-Off Between the Present and the Future (cont'd)

Observations

- An increase in capital goods at present will lead to a higher rate of economic growth in the future
- In the future, the economic system can produce more consumption goods

Comparative Advantage and Trade

Specialization

- Organization of economic activity among different individuals and regions
- Leads to greater productivity

Comparative Advantage and Trade

Absolute Advantage

- The ability to produce more units of a good or service using a given quantity of labor or resource inputs
- Equivalently, the ability to produce the same quantity of a good or service using fewer units of labor or resource inputs

Comparative Advantage and Trade

Comparative Advantage

- The ability to produce a good or service at a lower *opportunity cost*
- Is always a *relative* concept

Comparative Advantage and Trade

Absolute and Comparative Advantage

- A country or an individual **CAN** have an absolute advantage in the production of every good.
- However, by definition, a country or an individual **CANNOT** have a comparative advantage in the production of every good.

Comparative Advantage and Trade

Rational individuals choose their **comparative** advantage and then specialize

Specialization leads to division of labor

Adam Smith, in *The Wealth of Nations*, illustrated division of labor in pin making

Comparative Advantage and Trade

Division of Labor

- The segregation of resources into different specific tasks
- For example, in automobile production, one worker puts on bumpers, another worker puts on doors, and so on

Allocation of players to teams

The problem:

- You are in charge of allocating residents to your dormitory's baseball and basketball teams.
- You are down to the last four people, two of whom must be allocated to baseball and two to basketball.
- The following table gives each person's batting average and free-throw average

Allocation of players to teams

Name	Batting average	Free-throw average
Eric	50%	50%
Stan	70%	80%
Kyle	20%	40%
Leopold	10%	60%

Comparative Advantage and Trade

Analysis of absolute advantage, comparative advantage, and specialization is applicable to individuals, groups of people, or nations

As a result, *interstate* trade occurs in the United States and *international* trade occurs between nations

Comparative Advantage and Trade

When nations specialize where they have a comparative advantage and then trade with the rest of the world

- Economic efficiency improves
- Output increases
- Average standard of living rises

Comparative Advantage

Example:

- Suppose countries A and B produce only commodities X and Y.
- To simplify assume that the opportunity cost of production is constant in both countries.

Comparative Advantage

Example:

- The possibilities of production in each country are characterized by the following combinations:
- In country A: $(Y=1500, X=0)$ and $(Y=0, X=3)$
- In country B: $(Y=3000, X=0)$ and $(Y=0, X=1.5)$

Comparative Advantage

Example:

In autarky (no trade):

- Country A produces and consumes ($Y=750$, $X=1.5$)
- Country B produces and consumes ($Y=1000$, $X=1$)

Comparative Advantage

Example:

- Draw the PPC in each country.
- Determine the pattern of absolute advantage.
- Determine the pattern of comparative advantage.
- Explain how countries could benefit from specialization and trade.

END OF CHAPTER 2!
